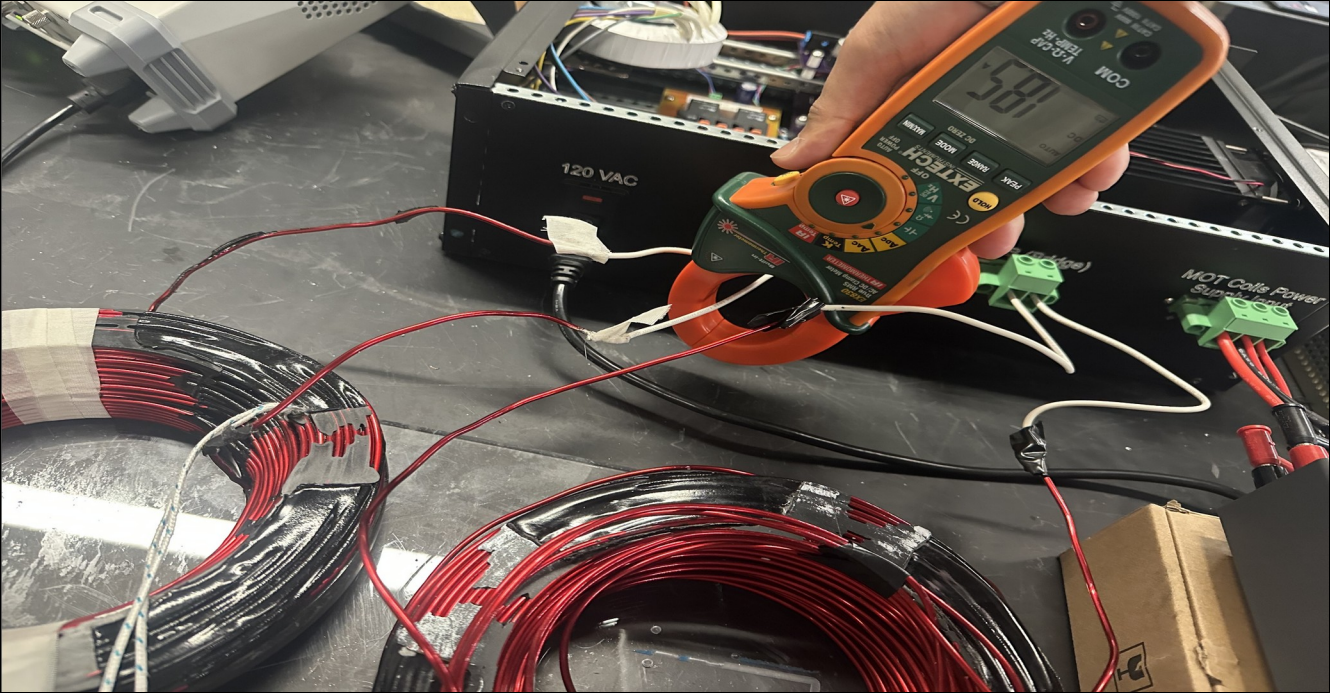


These Tests where done using a + and – 15V supplies.

1. The Setpoint and Monitor Voltage Relationship with Col Current

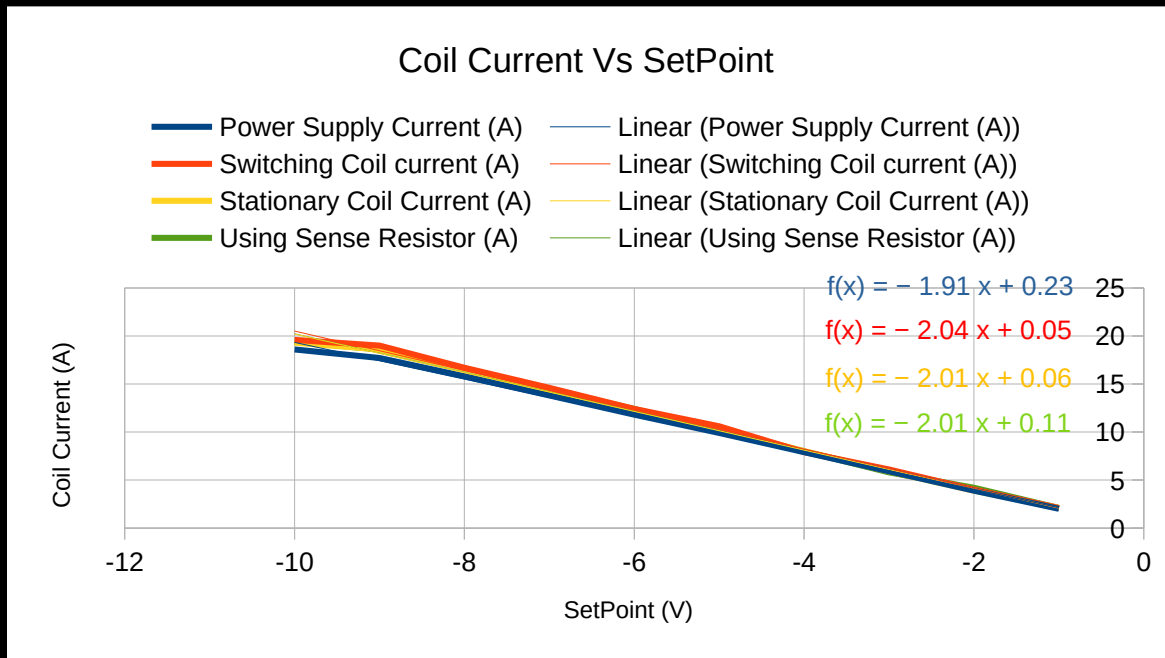
- For this part I Measured the Current in the coils using the Clamp On Ammeter as such:



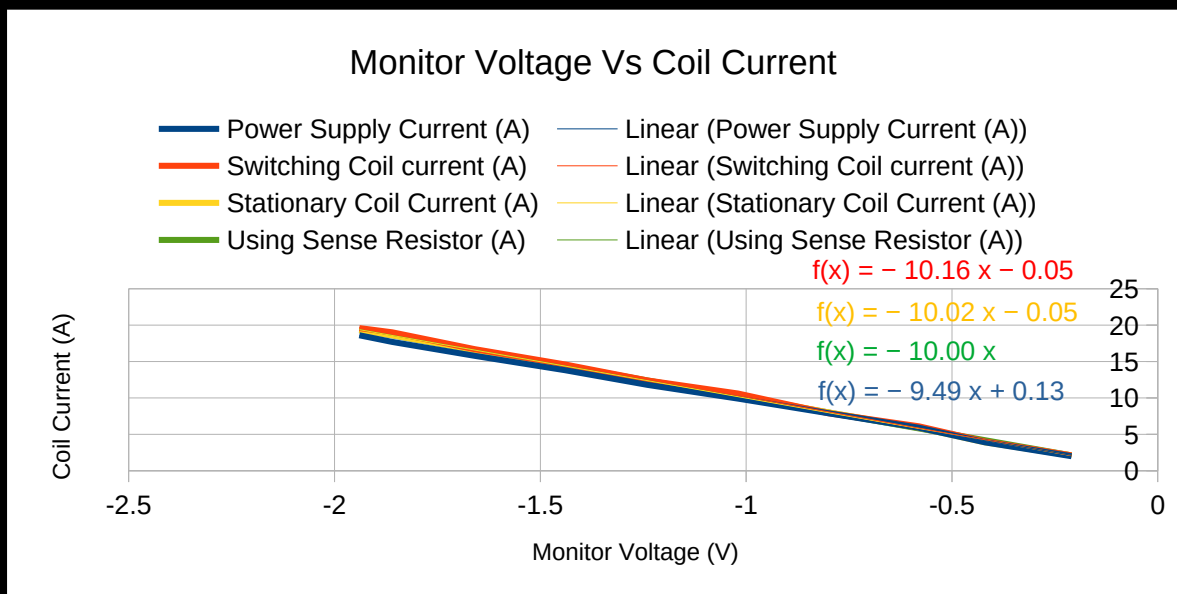
- Here is the data collected:

			Tolerance		
Sense Resistor	0.01 Ohms	+/- 0.1%			
SetPoint Voltage (V)	Monitor Voltage (V)	Power Supply Current (A)	Switching Coil current (A)	Stationary Coil Current (A)	Calc \w Sense Resistor (A)
-1	0.21	2	2.1	2.1	2.1
-2	0.42	3.9	4	3.9	4.2
-3	0.58	5.9	6.1	5.9	5.8
-4	0.8	7.9	7.9	8	8
-5	1.02	9.9	10.6	10.3	10.2
-6	1.24	11.8	12.4	12.3	12.4
-7	1.44	13.8	14.6	14.4	14.4
-8	1.66	15.8	16.7	16.6	16.6
-9	1.86	17.7	19	18.6	18.6
-10	1.94	18.6	19.6	19.3	19.4

- The Relationship With SetPoint and Coil Current



- The Relationship with Monitor Voltage and Current:



2. Rise and Fall Timing

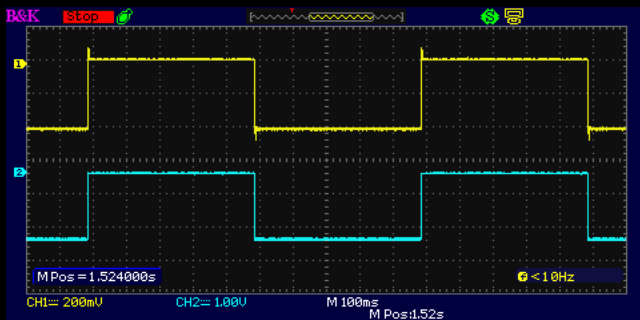
Note: The Rise time is when the set point voltage becomes negative and the Fall time is when the Setpoint Voltage becomes 0.

- The blue trace is the SetPoint and the Yellow trace is the monitor.

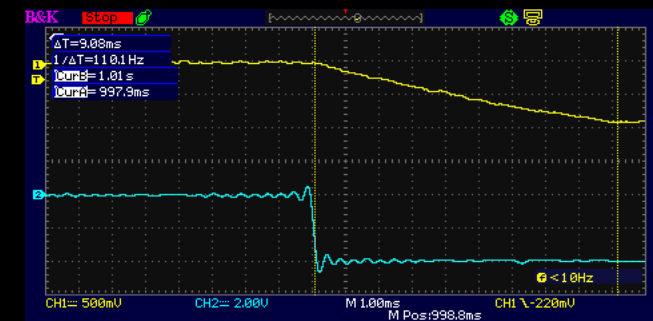
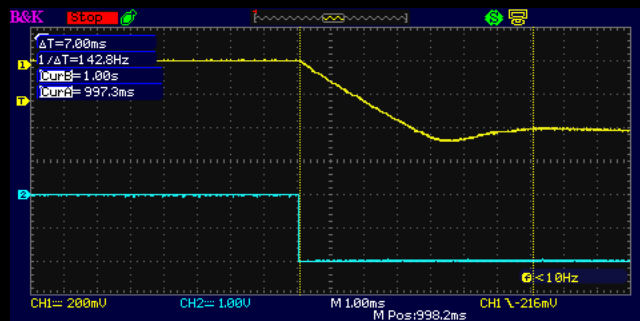
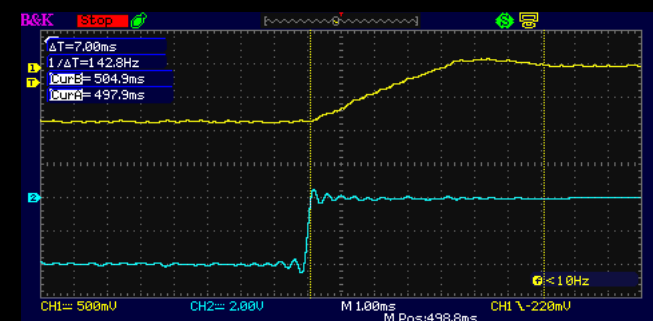
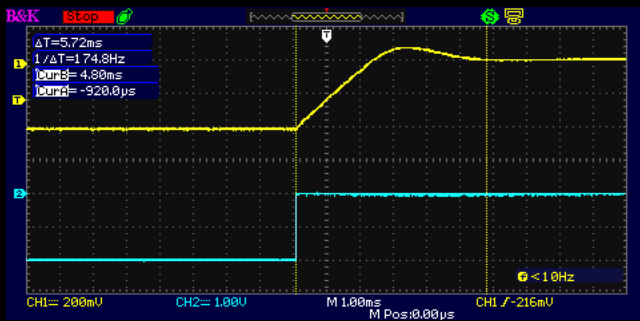
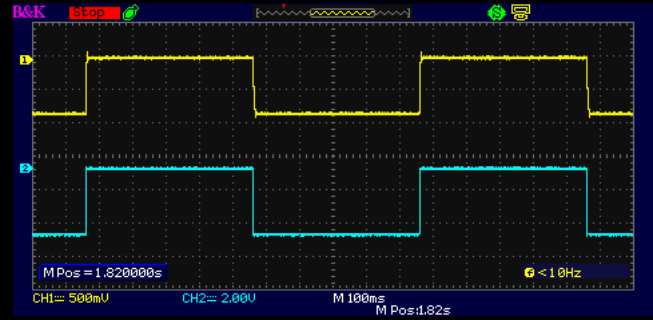
- Here is the Data Collected:

SetPoint Voltage (V)	RiseTime (ms)	FallTime (ms)
-2	7.00	5.72
-4	9.08	7.00
-6	18.40	10.16
-8	30.30	11.00
-10	67.60	12.12

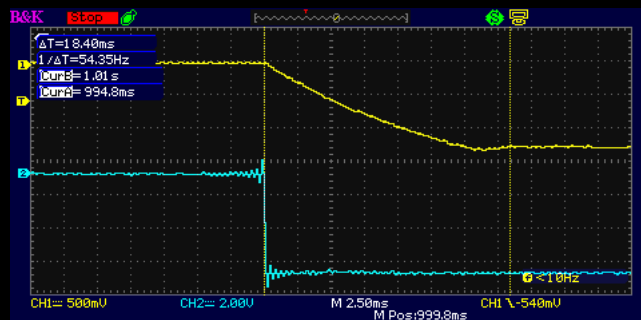
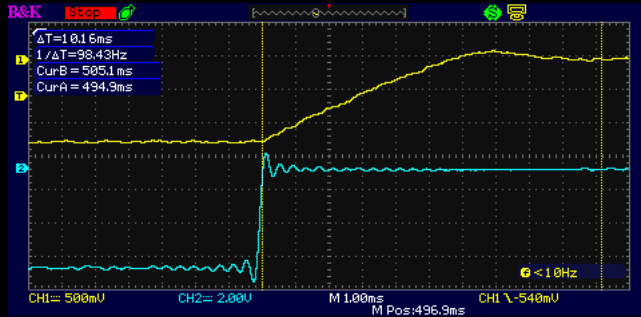
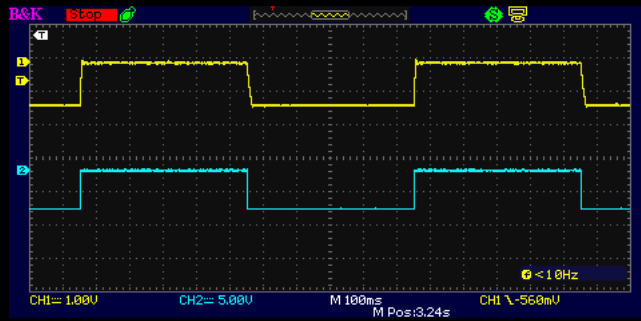
- At SetPoint -2V



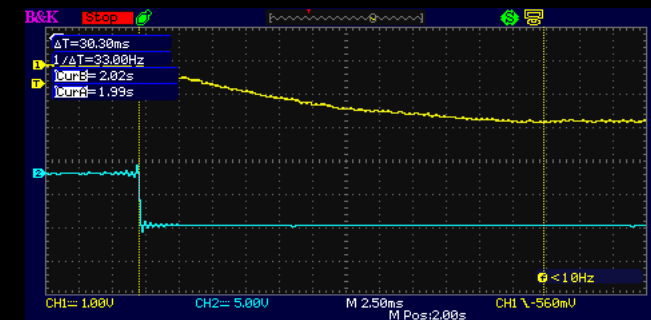
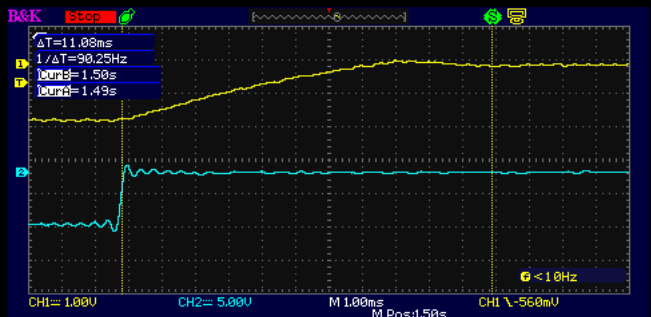
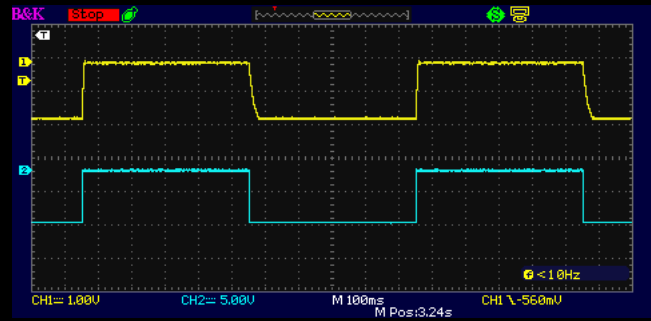
- At SetPoint -4V



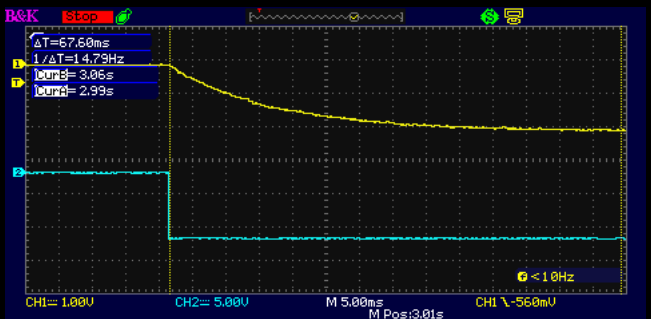
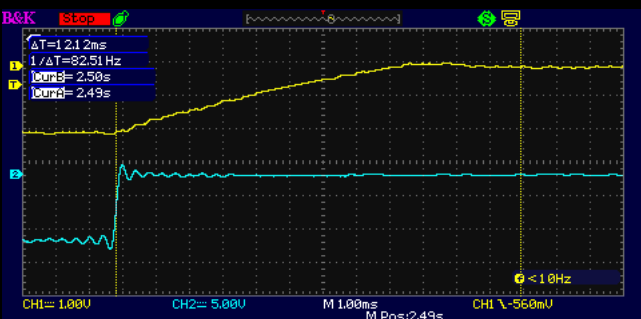
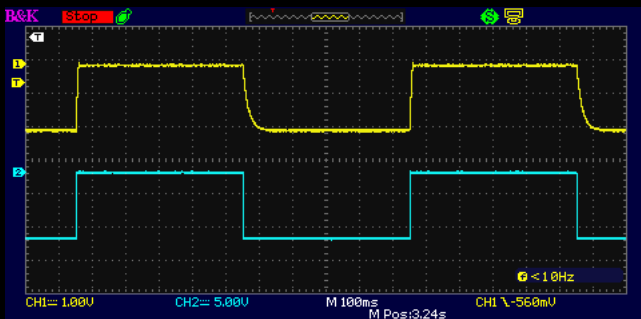
- At SetPoint -6V



- At SetPoint -8V



SetPoint -10V



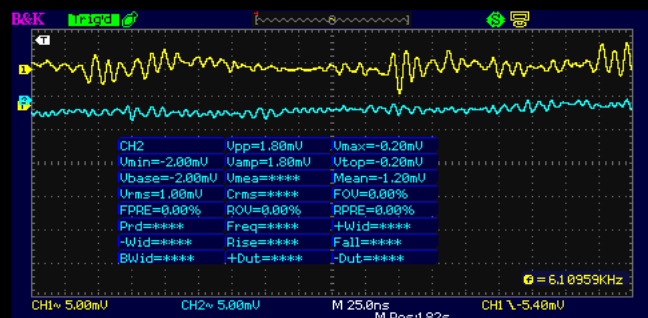
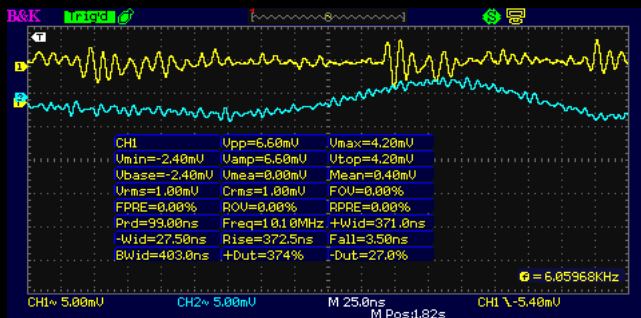
3. Protection Mechanism

In the video, I show how the protection mechanism is working. First, the function generator channel 1 is set to output a DC signal at -6V. Then I set channel 2 to output a square wave TTL with a high level 3.3V and low level at 0V. Once I turn channel 2 on, you can see that the protection LED are two different colors and it is flashing because of the square wave TTL. This means that the current is not allowed to switch right now because there is ~ 12A flowing through the coils right now. However, when I disconnect the Setpoint the current through the coils drops to 0. The relays start to click and you can see that the protection LED's are now both the same color. Link: <https://youtu.be/TYasHYS4kVA>

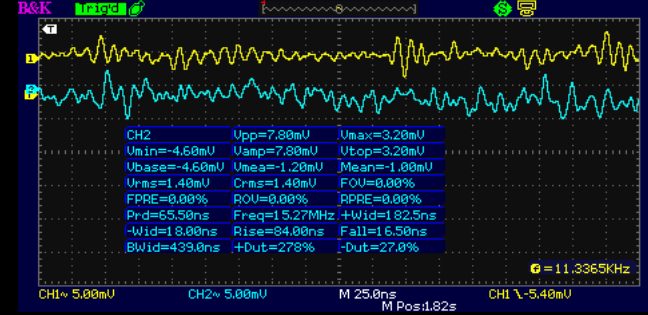
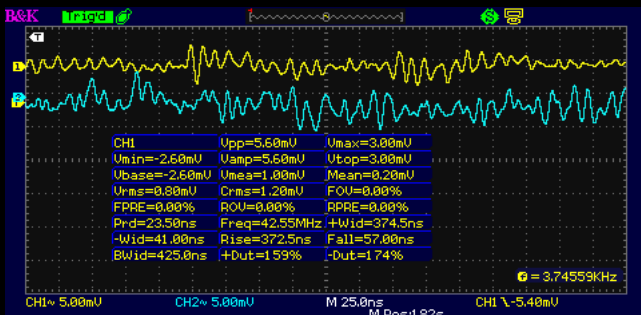
4. Noise From Monitor

Note: These measurements might not be correct, because I didn't refresh the scope after applying a new Setpoint. To see the noise I applied a DC setpoint, then I set both channels to AC coupling and zoomed into the signal.

- When no setpoint is provided



- At SetPoint -4V



- At SetPoint -8V

